Writing and Interpreting Circle Equations
Answer Key

High-School Geometry — Circles

Quick Answers

1. 7, 7
2. 36
3. 5, 25
4. 4.47
5. $(x-3)^2+(y+4)^2-36$, 6
6. -2, 6
7. $(x+5)^2+(y-2)^2-9$, 3
8. -7

Curriculum

Level: High-School Geometry — Circles
HSG-C.B.5 – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–5 of 5 across 13 tagged checks.

Common wrong answers

4. *If they got 10: halved the constant instead of taking its square root*
5. *If they got 3.32: took the square root of the original constant term instead of completing the square first*
8. *If they got 33: added the constant term instead of subtracting it when completing the square*

1. Radar: $(x-4)^2+(y+3)^2 = 49$**, in miles.** (a) Match the equation to $(x-h)^2+(y-k)^2 = r^2$. Here $x-4$ gives $h=4$, and $y+3$ is $y-(-3)$, so $k=-3$. The tower is at $(4,-3)$; the sign flips because the form subtracts the center's coordinates.
(b) The right-hand side is r^2, not r, so $r=\sqrt{49}=7$. Radius: $\boxed{7 \text{ mi}}$.
(c) The plane at $(11,-3)$ is on the edge exactly when its distance from the center is the radius:

$$\sqrt{(11-4)^2+(-3-(-3))^2} = \sqrt{49+0} = 7$$

so the distance is $\boxed{7 \text{ mi}}$, which equals the radius — the plane is exactly on the edge of the range.

2. Sprinkler at $(-2, 5)$, reach 6 meters. The watered boundary is every point exactly 6 meters from the sprinkler, so the center is $(-2, 5)$ and $r = 6$. Substituting into $(x-h)^2 + (y-k)^2 = r^2$ and remembering that $h = -2$ makes $x - (-2) = x + 2$:

$$(x+2)^2 + (y-5)^2 = 6^2$$

The right-hand side is $r^2 = \boxed{36}$, so the equation is

$$\boxed{(x+2)^2 + (y-5)^2 = 36}$$

3. Cell tower at $(3, -2)$, reaching the town at $(8, -2)$. (a) The radius is the distance from the tower to the town:

$$r = \sqrt{(8-3)^2 + (-2-(-2))^2} = \sqrt{25+0} = 5$$

Radius: $\boxed{5 \text{ km}}$.
(b) Now write the equation with center $(3, -2)$ and $r = 5$, so $r^2 = 25$:

$$\boxed{(x-3)^2 + (y+2)^2 = 25}$$

Check the town: $(8-3)^2 + (-2+2)^2 = 25 + 0 = 25$, so $(8, -2)$ is on the boundary exactly as the story says.

4. Ripple: $x^2 + (y-6)^2 = 20$, in meters. Write x^2 as $(x-0)^2$, so the center is $(0, 6)$. The right side is $r^2 = 20$, so

$$r = \sqrt{20} = 2\sqrt{5} = 4.4721\ldots$$

Rounded to two decimal places, the radius is $\boxed{4.47 \text{ m}}$. A common error is to halve 20 and report 10; the right side is a square, so it needs a square root, not a half.

5. Running track: $x^2 + y^2 - 6x + 8y - 11 = 0$, in meters. Group the x-terms and the y-terms and move the constant across:

$$(x^2 - 6x) + (y^2 + 8y) = 11$$

Complete each square: half of -6 is -3, and $(-3)^2 = 9$; half of 8 is 4, and $4^2 = 16$. Add both to *both* sides:

$$(x^2 - 6x + 9) + (y^2 + 8y + 16) = 11 + 9 + 16$$

$$\boxed{(x-3)^2 + (y+4)^2 = 36}$$

So the center is $(3, -4)$ and $r = \sqrt{36} = \boxed{6 \text{ m}}$. Note the added constants must go on the right too — taking $\sqrt{11} \approx 3.32$ from the original constant is the standard mistake here.

6. Fence $(x-2)^2+(y-1)^2 = 25$ meets the path $y = 4$. Substitute $y = 4$ into the circle equation:

$$(x-2)^2 + (4-1)^2 = 25 \quad \Longrightarrow \quad (x-2)^2 + 9 = 25$$
$$(x-2)^2 = 16 \quad \Longrightarrow \quad x - 2 = \pm 4$$

so $x = 2 + 4 = 6$ or $x = 2 - 4 = -2$:

$$\boxed{x = -2 \quad \text{or} \quad x = 6}$$

Both points, $(-2, 4)$ and $(6, 4)$, are 5 meters from the center, as they must be.

7. Rewrite $x^2 + y^2 + 10x - 4y + 20 = 0$. Group and move the constant:

$$(x^2 + 10x) + (y^2 - 4y) = -20$$

Half of 10 is 5 and $5^2 = 25$; half of -4 is -2 and $(-2)^2 = 4$. Add both to each side:

$$(x^2 + 10x + 25) + (y^2 - 4y + 4) = -20 + 25 + 4$$

$$\boxed{(x+5)^2 + (y-2)^2 = 9}$$

The center is $(-5, 2)$ and the radius is $\sqrt{9} = \boxed{3 \text{ units}}$.

8. The classmate's equation $x^2 + y^2 + 4x + 6y + 20 = 0$. (a) Group and complete both squares: half of 4 is 2 with $2^2 = 4$, and half of 6 is 3 with $3^2 = 9$, so

$$(x^2 + 4x + 4) + (y^2 + 6y + 9) = -20 + 4 + 9$$

$$(x+2)^2 + (y+3)^2 = 4 + 9 - 20$$

The right-hand side is $\boxed{-7}$.
(b) **Open explanation — grade against this model.** A sum of two squares can never be negative, because each square is zero or positive for every real x and y. So no point (x, y) in the plane satisfies the equation: its graph is empty, not a circle. The classmate is wrong twice — there is no circle at all here, and even when the right-hand side is positive it equals r^2, so a right side of 20 would mean $r = \sqrt{20}$, not $r = 20$. Accept any answer that says a negative right-hand side means no real graph and explains why (squares cannot sum to a negative).